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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY

Paper 1 Multiple Choice Questions

8875/01

30 September 2014

1 hour

Additional Materials: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A MCQ [30 marks]

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

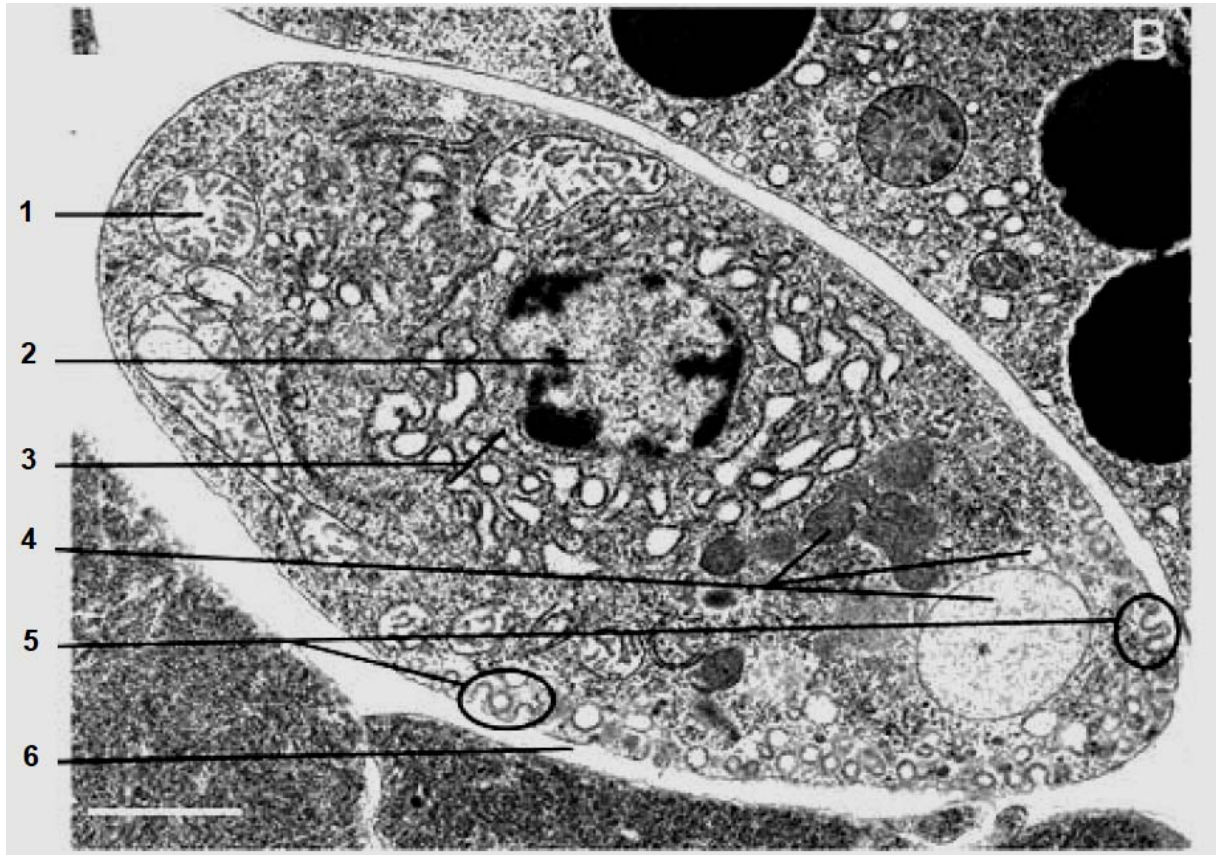
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **20** printed pages and **0** blank page.

Answer **all** questions in this section.

1



Which of the structure(s) in the above electron micrograph is/are found abundantly in actively respiring tissues?

- A 1 only
- B 4 only
- C 2 and 5 only
- D 3 and 6 only

2 Which of the following organelles are DNA present?

- A Chloroplasts and ribosomes
- B Chloroplasts and mitochondria
- C Mitochondria and lysosomes
- D Nucleoli and ribosomes

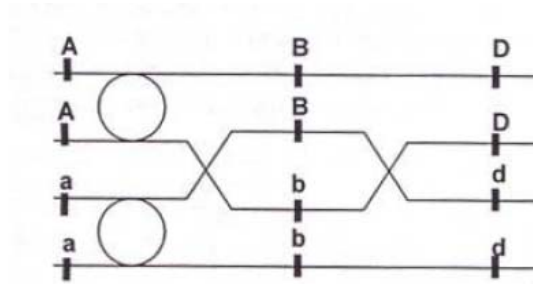
- 3 Some germinating seeds were crushed with water and the extract was tested. The table below shows the results of these tests.

Test	Result
Barford's test for disaccharides	Positive
Benedict's test	Yellow precipitate
Biuret test	Purple colour
Clinistix test for glucose	Negative
Emulsion test	Clear solution
Iodine test	Blue-black colour

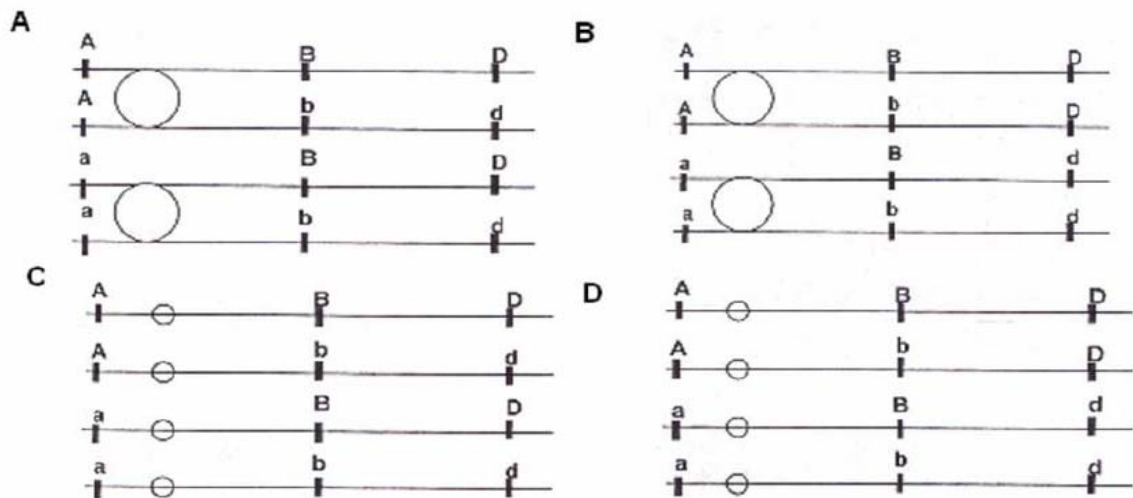
Which molecules were present in the extract?

- A Fat, maltose and starch only.
- B Glucose, protein and starch only.
- C Maltose, protein and starch only.
- D Sucrose, protein and starch only.

4



The diagram above shows a bivalent. Which of the following correctly represents the final products at the end of Meiosis II?



- 5 Each polypeptide chain of haemoglobin contains a number of regions in the form of an alpha helix.

Which feature of the haemoglobin molecule is responsible for this?

- A Disulphide bonds between the four polypeptide chains at the quaternary structure.
- B Hydrogen bonds between R groups of the amino acids in the secondary structure.
- C Hydrogen bonds within the polypeptide chain in the secondary structure.
- D Hydrophobic interactions within the polypeptide chain in the tertiary structure.

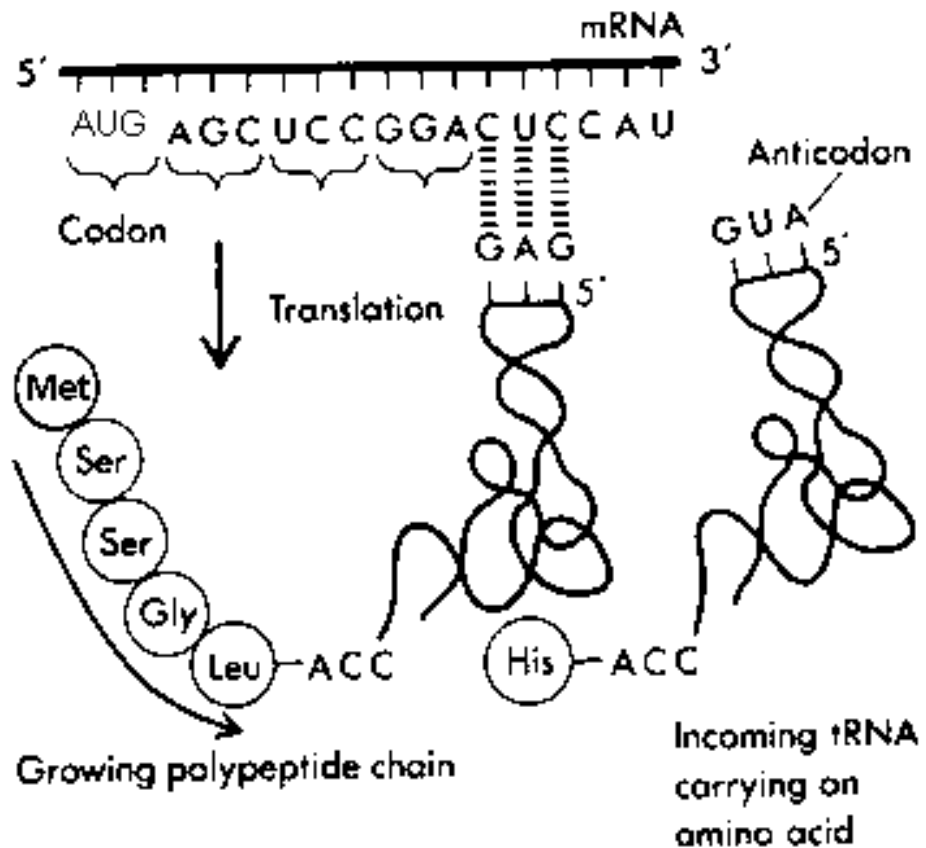
- 6 An experiment is carried out to show that DNA replication is semi-conservative.

Escherichia coli cells whose DNA contains only 'heavy' nitrogen (^{15}N) are grown in a culture medium containing the normal isotope of nitrogen (^{14}N) and allowed to divide three times.

What will be the percentage of DNA molecules containing both forms of nitrogen after the third division?

- A 12.5%
 - B 25%
 - C 50%
 - D 100%
- 7 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?
- I The anti-parallel arrangement of the DNA strands.
 - II The RNA primers are required to initiate DNA elongation.
 - III The template strands must be read from the 3' to 5' direction
 - IV Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.
- A I and III
 - B I and IV
 - C II and III
 - D II and IV

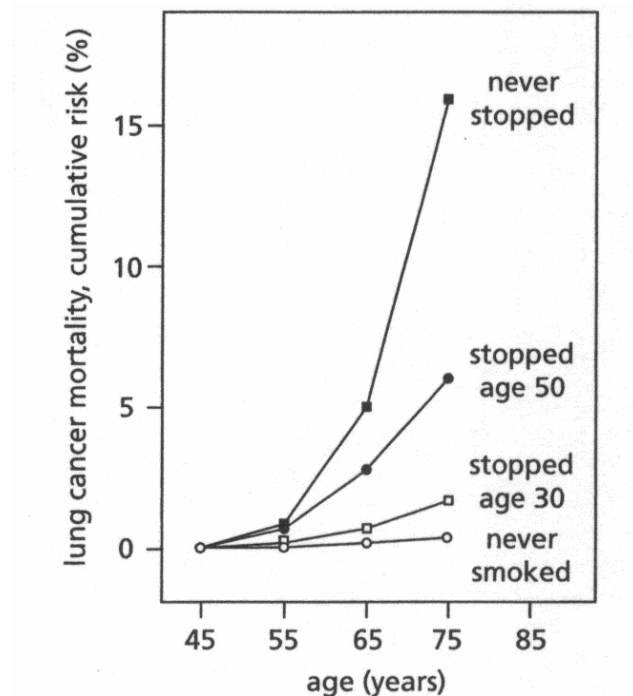
- 8 The diagram below shows the process of translation.



Which of the following are **CORRECT**?

- I The ribosome is translocating from right to left.
 - II The diagram shows degeneracy of the genetic code.
 - III The polypeptide would be attached to the tRNA after histidine was added.
 - IV The number of hydrogen bonds formed between the respective codons and anticodons of the two tRNAs are equal.
- A I and IV
 B I and II
 C II and III
 D II and IV

- 9 Mortality due to lung cancer was followed in groups of males in the United Kingdom for 50 years. The cumulative risk of dying from lung cancer as a function of age and smoking habits for four groups of males is shown in the figure below.



Which of the following best explains the trends observed in the graph?

- I The slower the rate of accumulation of mutations, the lower the cumulative risk.
 - II When an individual stops smoking, he will undergo a decreased rate of mutation accumulation.
 - III The presence of cigarette smoke will cause the accumulation of mutations to occur at an increased rate.
 - IV There is little risk of a non-smoker dying of lung cancer.
- A I and IV only
 B II and III only
 C III and IV only
 D All of the above

- 10** The table shows the blood group phenotypes resulting from crosses between different genotypes.

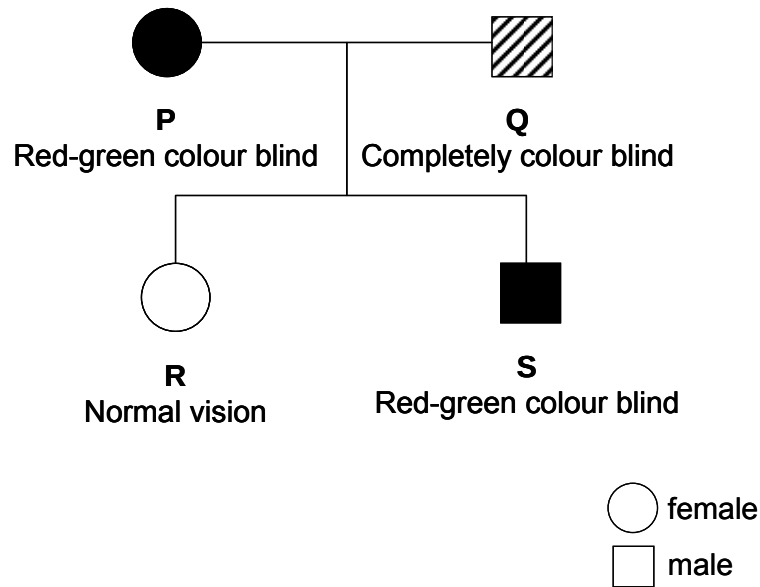
Genotype of 1 st parent	Genotype of 2 nd parent	
	$I^A I^A$	$I^A I^O$
$I^A I^O$	A	A and O
$I^B I^O$	A and AB	A, B, O and AB

Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria. What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

- A** 1 in 16
- B** 1 in 8
- C** 3 in 16
- D** 3 in 8

- 11** Red-green colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**. Complete colour blindness is controlled by a different gene which is not on the X chromosome. The allele for the development of normal cones (pigment cells in the retinal layer of the eye), **B**, is dominant to the allele for no cone development, **b**.

The figure below shows the phenotypes of members of a different family in which both types of colour blindness occur.



Which of the following are possible genotypes for individuals **P**, **Q** and **S**?

	P	Q	S
A	BBX^GX^G	bbX^GY	BBX^gY
B	BbX^GX^g	BbX^gY	BbX^GY
C	BBX^gX^g	bbX^GY	BbX^gY
D	BbX^GX^g	BbX^gY	BBX^GY

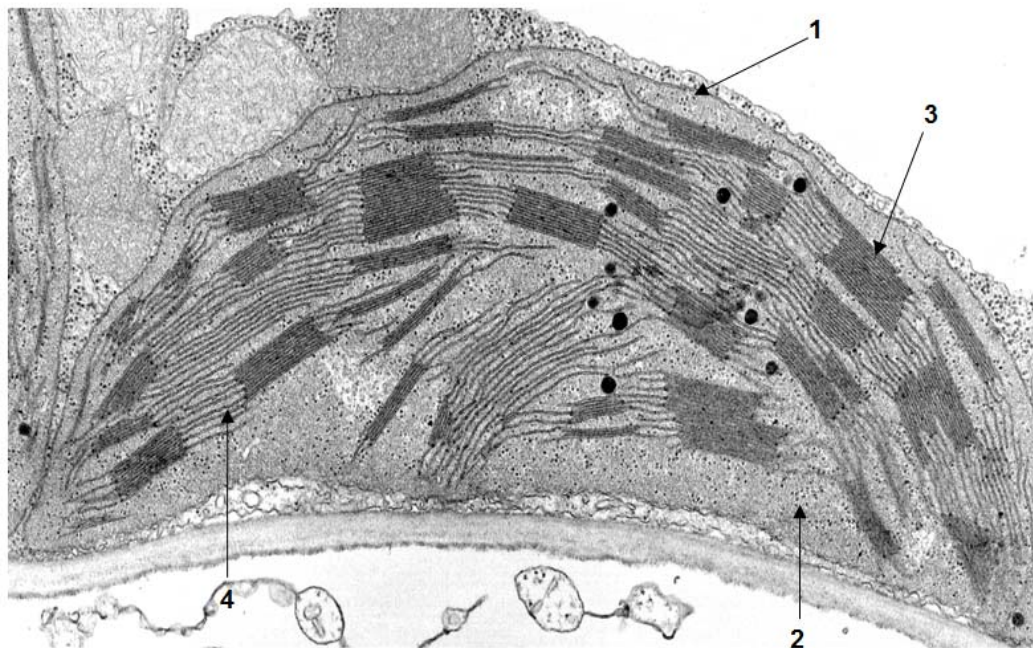
- 12** Four alleles determine coat colour in cats. The **C** allele codes for grey colour and is dominant over the **c^b** and **c^s** alleles. Both the **c^b** and **c^s** alleles are codominant and code for sepia colour. However, the **c^b** allele is also temperature sensitive and codes for black colour if during the fetal development stage, the cat is exposed to cool temperatures. Both the **c^b** and **c^s** alleles are dominant over the **c** allele which results in an albino.

Which of the following crosses could produce offsprings that have sepia colour with black patches?

- | | | | |
|------------|------------------|---|---------------------------------|
| I | grey colour cat | x | grey colour cat |
| II | sepia colour cat | x | sepia colour with black patches |
| III | grey colour cat | x | sepia colour cat |
| IV | sepia colour cat | x | sepia colour cat |

- A** I, II and III
- B** II and III
- C** II, III and IV
- D** All of the above

- 13 Which of the following is correctly associated with the function or characteristic of the labelled structure?

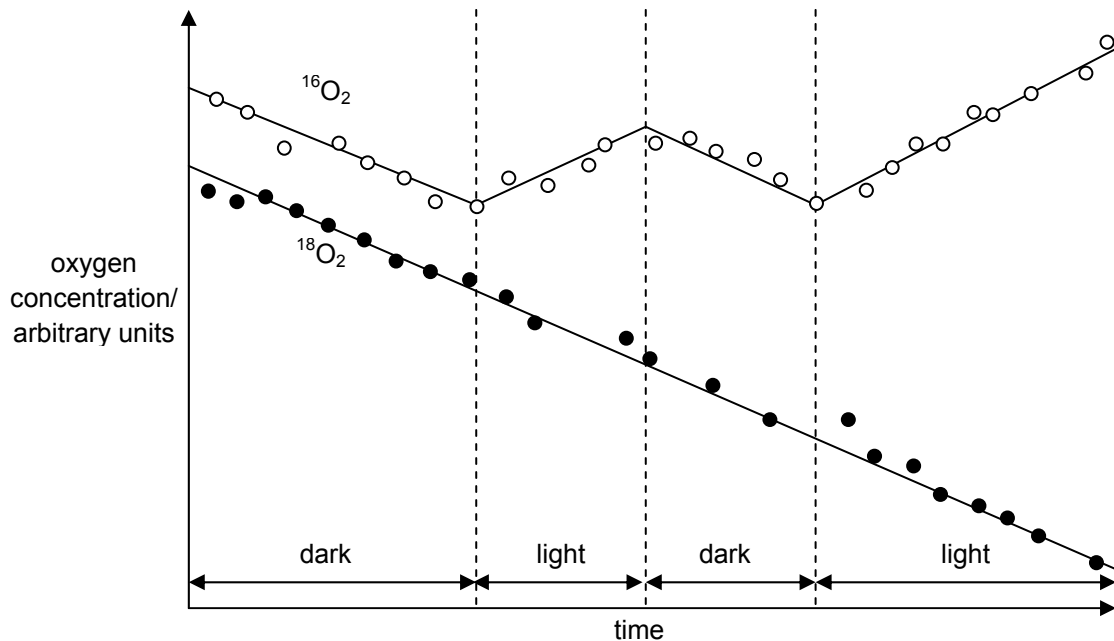


	help to maintain high H^+ concentration	made up of phospholipid	has stalked particles embedded in it	where ATP is synthesised
A	1 and 2	1, 2, 3 and 4	4	2
B	3 and 4	2 and 4	2 and 3	1 and 2
C	1 and 4	1, 2 and 3	1 and 4	2 and 4
D	3 and 4	1, 3 and 4	3 and 4	2

- 14 Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

A mixture of oxygen isotopes $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to $^{16}\text{O}_2$ only. During the following hour, changes in concentration of these gases in the suspension were measured in light and dark conditions.

The graph below shows the results.



What caused the concentration of $^{16}\text{O}_2$ to rise in light?

- A H_2^{18}O was being photolysed more rapidly than H_2^{16}O .
- B $^{16}\text{O}_2$ was being absorbed at different rates in light and dark.
- C $^{16}\text{O}_2$ was being produced in photosynthesis but was not being absorbed in respiration.
- D $^{16}\text{O}_2$ was being produced in photosynthesis faster than it was being absorbed in respiration.

- 15** $^{14}\text{CO}_2$ is added to a suspension of photosynthesizing green algae. Which compound will be labeled first with ^{14}C ?
- A** Glucose
 - B** Triose phosphate
 - C** Glycerate-3-phosphate
 - D** Ribulose biphosphate

- 16** During aerobic respiration ATP can be formed by oxidative phosphorylation in the electron transport system and by glycolysis.

In the complete oxidation of two molecules of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

- A** 40%
 - B** 45%
 - C** 80%
 - D** 90%
- 17** Steven has been unable to exert much energy. He goes to a doctor for help and is sent to the hospital for some tests. It was discovered that his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal.

Which of the following is the best explanation for his condition?

- A** His mitochondria lack the transport protein that moves pyruvate across the inner mitochondrial membrane.
- B** His cells cannot move NADH from glycolysis into the mitochondria.
- C** His cells contain an inhibitor that prevents oxygen use in his mitochondria.
- D** His cells have a defective electron transport chain, so glucose is reduced to lactate instead of to acetyl CoA.

18 Six tubes were set up as shown in the table below.

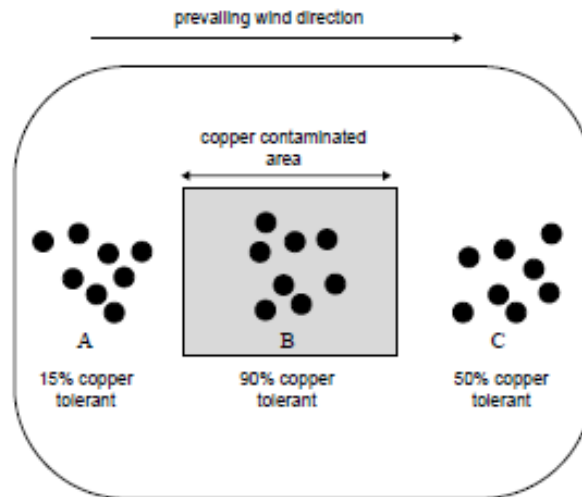
tube	Contents
1	Glucose + homogenized plant cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells
5	Pyruvate + mitochondria
6	Pyruvate + cytoplasm lacking organelles

After incubation, each sample was analysed to determine the presence of carbon dioxide and ethanol.

In which tubes are lactate most likely to be present?

- A** 1 and 3 only
- B** 2, 3, 5 and 6 only
- C** 4, 5, and 6 only
- D** 3 and 6 only

- 19 The figure below shows an aerial view of a region which included a section of copper contaminated soil (shaded grey on the diagram). The black circles represent populations of grass growing in areas A, B and C. The figure also includes the percentage of copper resistant plants in areas A, B and C. Tolerance to copper in the soil is genetically determined.



It is reasonable to conclude from the information that _____ .

- A all plants in the copper contaminated area would be homozygous at the gene locus for copper tolerance
 - B high levels of copper tolerance would be a selective advantage for all plants in areas A, B and C
 - C the difference in copper tolerance between populations in areas A and C is the result of mutation
 - D gene flow is occurring between populations in areas B and C
- 20 When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than houseflies in Britain.

What could account for the difference in the rates of the development of resistance?

- A More insecticides were used in Britain.
- B More insecticides were used in malarial regions.
- C Mosquitoes have fewer random mutations when exposed to insecticides.
- D Mosquitoes have more random mutations when exposed to insecticides.

- 21** A new species of organism is considered to have evolved when its population _____ .
- A** is isolated from the rest of the population by a geographical barrier.
 - B** shows increased variation due to mutations.
 - C** can no longer interbreed successfully with the rest of the population.
 - D** Is subjected to increased selection pressure in its habitat.
- 22** Which one of the following statements is related to the principle of adaptive radiation in evolution?
- A** Unrelated organisms living in similar environments may tend to resemble each other.
 - B** Related organisms living in different environments show modifications of an underlying unity of pattern.
 - C** Selection in a constant environment tends to maintain the most frequently occurring variety of a species.
 - D** Species evolve by selection altering the gene frequencies in interbreeding populations.
- 23** Which of the following poses the greatest limitation of the Human Genome Project in disease analysis?
- A** It is not possible to store the genomic information of all individuals.
 - B** There is too large a genetic difference between individuals to determine the loci of genes of interest.
 - C** Too few model organisms exist for comparative genomic studies to take place.
 - D** Environmental factors and gene interactions in disease are not taken into account.

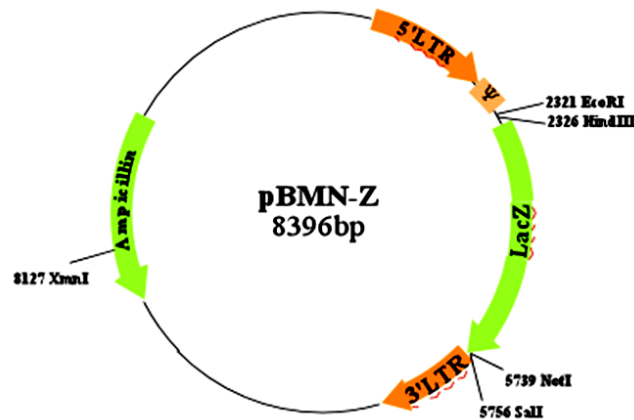
- 24 A sample of DNA was treated separately with two different restriction enzymes. The products of digestion was then run on a gel and stained with ethidium bromide.

Which of the following statement(s) **CANNOT** account for the difference in the *number* of bands seen in the gel?

- I The two enzymes have active sites that are of different shapes.
- II One of the enzymes produces blunt ends and the other sticky ends.
- III The position of the restriction sites for each restriction enzyme differs
- IV Ethidium bromide binds specifically to only certain DNA fragments.

- A II, III and IV only
- B I and III only
- C II and IV only
- D II only

- 25 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the MAL protein into the pBN-Z vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for MAL protein were to be inserted into Lac Z site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	Chemicals to be added		Colour of colonies
A	Ampicilin	X-gal	Blue
B	β -galactosidase	X-gal	Blue
C	Ampicilin	X-gal	White
D	β -galactosidase	lactose	White

- 26 Totipotency is demonstrated when _____.

- A mutations in genes result in the development of misplaced appendages.
- B a cell isolated from a plant leaf grows into a normal adult plant.
- C an embryonic cell divides and differentiates.
- D the replacement of the nucleus of an unfertilized egg with that of an intestinal cell converts the egg to an intestinal cell.

- 27** What is the role of stem cells with regard to the function of adult tissues and organs?
- A** Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter cell that remains a stem cells and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.
 - B** Stem cells are embryonic cells that persist in the adult, and can give rise to all of the cell types in the body.
 - C** Stem cells are differentiated cells that have yet to express the genes and produce proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.
 - D** Stem cells are fully differentiated cells that reside under the surface of epithelial tissue, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
- 28** One type of genetically modified organism that is close to commercial consideration is a tilapia species which has a salmon gene coding for growth hormone. What benefit does this GM fish provide?
- A** Increasing the nutritional quality of food.
 - B** Increasing the yield of food.
 - C** Mass produce growth hormones for medical treatment in humans.
 - D** Mass produce growth hormones to speed up growth in salmon.
- 29** Some crop plants have been genetically modified to give resistance against specific insects. Which feature is least likely to be affected by this development?
- A** crop damage caused by specific insects
 - B** intital presence of the specific insects in the area of crop growth
 - C** population size of the specific insects
 - D** use of insecticides in the area of crop growth

- 30** What is **NOT** a possible issue of concern over the creation of genetically modified farmed animals?
- 1 Genetic engineering may result in the creation of new proteins that are harmful to the organisms that produce or consume them.
 - 2 Cross species gene transfer may compromise the genome integrity of the species involved.
 - 3 Over production of certain gene products may cause undue stress to the genetically modified farmed animals.
 - 4 Some genetically modified food products may not be acceptable to certain groups of people.
- A 1 and 4
- B 2 and 3
- C 1, 3 and 4
- D None of the above

END OF PAPER

2014 Y6 Preliminary Exam H1**MCQ Answer Scheme**

1	A	11	C	21	C
2	B	12	D	22	B
3	C	13	D	23	D
4	B	14	D	24	A
5	C	15	C	25	C
6	B	16	D	26	B
7	A	17	A	27	A
8	C	18	D	28	B
9	D	19	D	29	B
10	C	20	B	30	D